

HOW A CHAMPIONSHIP FRACTURED A FANDOM

Measuring McLaren's fan polarization through Reddit, 2024-2025.

THE BUSINESS OF FANDOM.

Formula 1 is a media and sponsorship business in which fan perception is brand value. When teams make controversial in-race decisions, that value is publicly negotiated — in real time — on social media.

- › Social platforms are the primary surface for fan sentiment in elite sports.
- › Traditional fan-perception tools (surveys, focus groups) are slow and expensive.
- › Reddit communities aggregate the most engaged segment of any fanbase.
- › Opportunity: real-time social listening as a polarization detector.

F1 INDUSTRY REVENUE

\$3.7B

2024 — driven primarily by broadcasting, sponsorship and race-promotion contracts.



THE SELECTED IDEA

When a sports brand makes **controversial in-season decisions**, can we detect from social media alone whether its fanbase has fractured — and pinpoint **when, where, and how severely?**

SUBJECT

McLaren F1, navigating an internal championship battle between teammates Lando Norris and Oscar Piastri across 2024-2025 — defined by ten controversial 'team orders' moments, starting with the Hungarian GP 2024 swap (patient zero).



THE DATASET.

SOURCE — REDDIT.COM/R/FORMULA1 · 2024-2025

VOLUME 01

5,213

Reddit posts collected from r/formula1 via the public JSON endpoint.

→ 44% FALL WITHIN THE 2024-2025 TARGET WINDOW

VOLUME 02

143k

Comments fetched with full author and parent-thread preserved.

→ 110,960 IN 2024-2025 WINDOW ANALYZED

PEOPLE

32k

Unique commenters — the input to community detection.

→ 8,329 ACTIVE USERS (≥3 POSTS)

ANCHORS

10

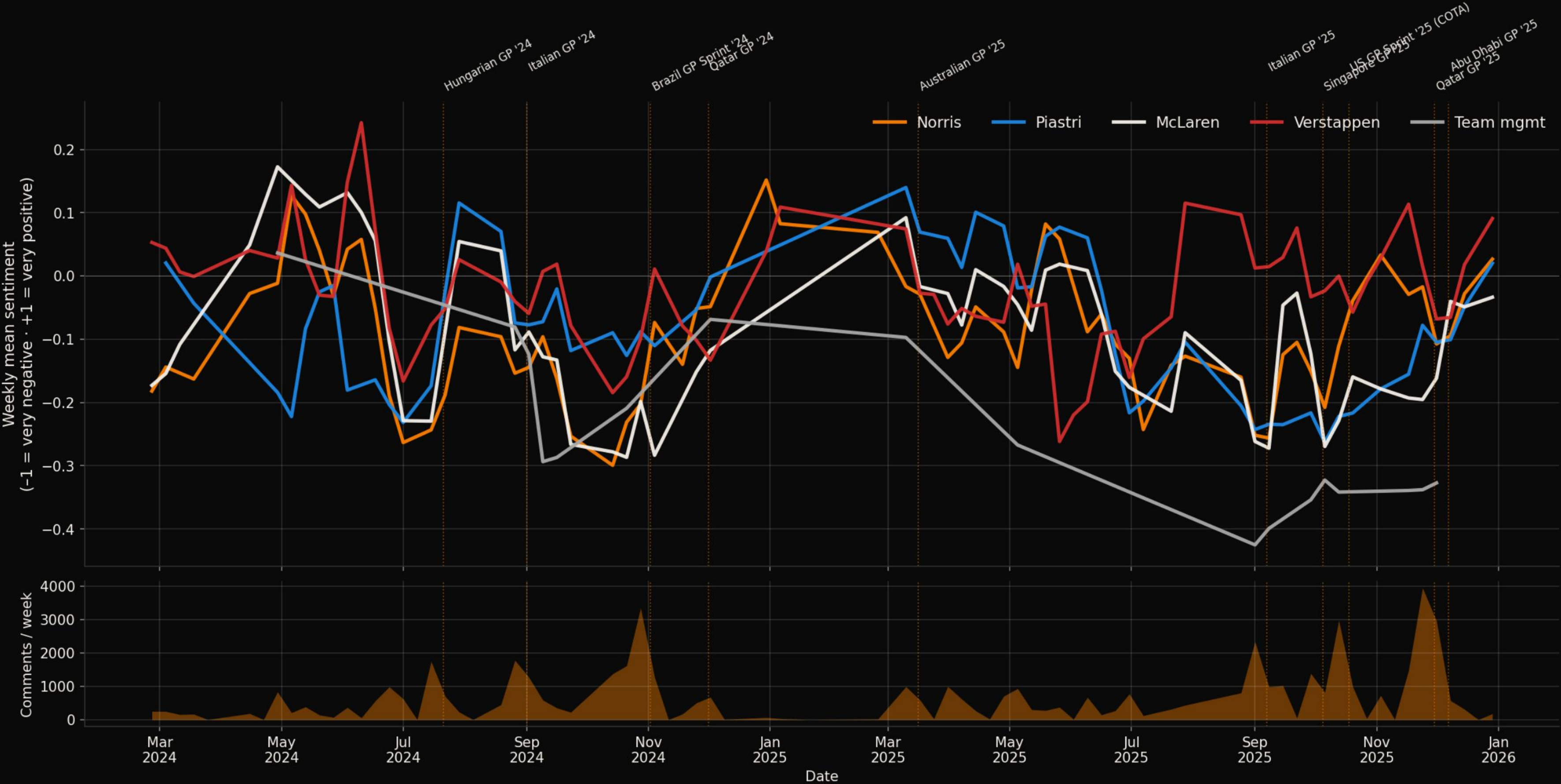
Flashpoint races identified as analytical anchors (F1-expert sourced).

→ PATIENT ZERO: HUNGARIAN GP 2024



DID SENTIMENT ACTUALLY CRASH?

ANALYSIS 1 · ASPECT-ORIENTED SENTIMENT · CARDIFFNLP/TWITTER-ROBERTA · WEEKLY · 110,960 COMMENTS



WHAT WERE FANS ARGUING ABOUT?

2025-09-07 • ITALIAN GP

"Slow pit stops are part of racing"

"We said a slow pit stop was part of racing. I don't really get what changed here." — Piastri's iconic radio message.

pit • stop • slow • swap • undercut

+0.00

2025-10-05 • SINGAPORE GP

"Papaya rules" emerges as a topic

LDA isolates the explicit "papaya rules" bigram as a topic-defining term — the framework itself is now being debated.

papaya • rules • contact • incident

+0.06

2025-10-18 • US GP SPRINT

Zak Brown enters the discourse

First and only event where 'Zak' is a top topic term. Fans name and blame leadership for the Norris-Piastri collision.

piastri • norris • zak

-0.02

2025-11-30 • QATAR GP

Strategy failure crystallized

Alternative-strategy decision sealed Verstappen's win. Profanity surfaces as defining vocabulary in the dominant topic.

lando • mclaren • pit • said

-0.01

2024-07-21 • HUNGARIAN GP

Patient zero of the schism

The original Norris-Piastri swap. Reluctance over the radio became the defining audio clip of the McLaren saga.

lando • mclaren • oscar • win

+0.08

2025-12-07 • ABU DHABI GP

Bittersweet championship topic

Highest-sentiment topic of the dataset (+0.24) — yet team-management sentiment hits a season low (-0.33) the same week.

lando • max • championship • won

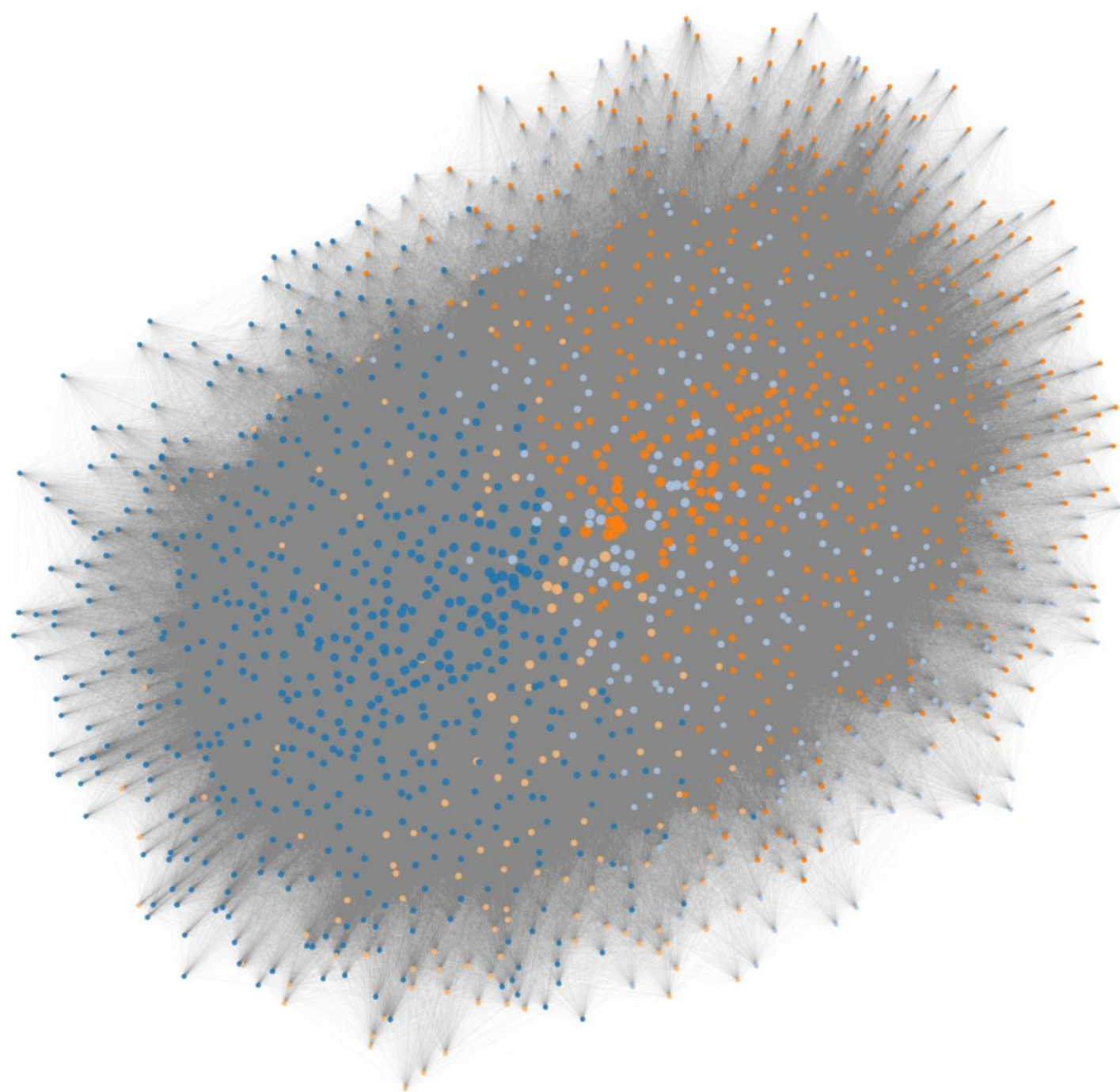
+0.24



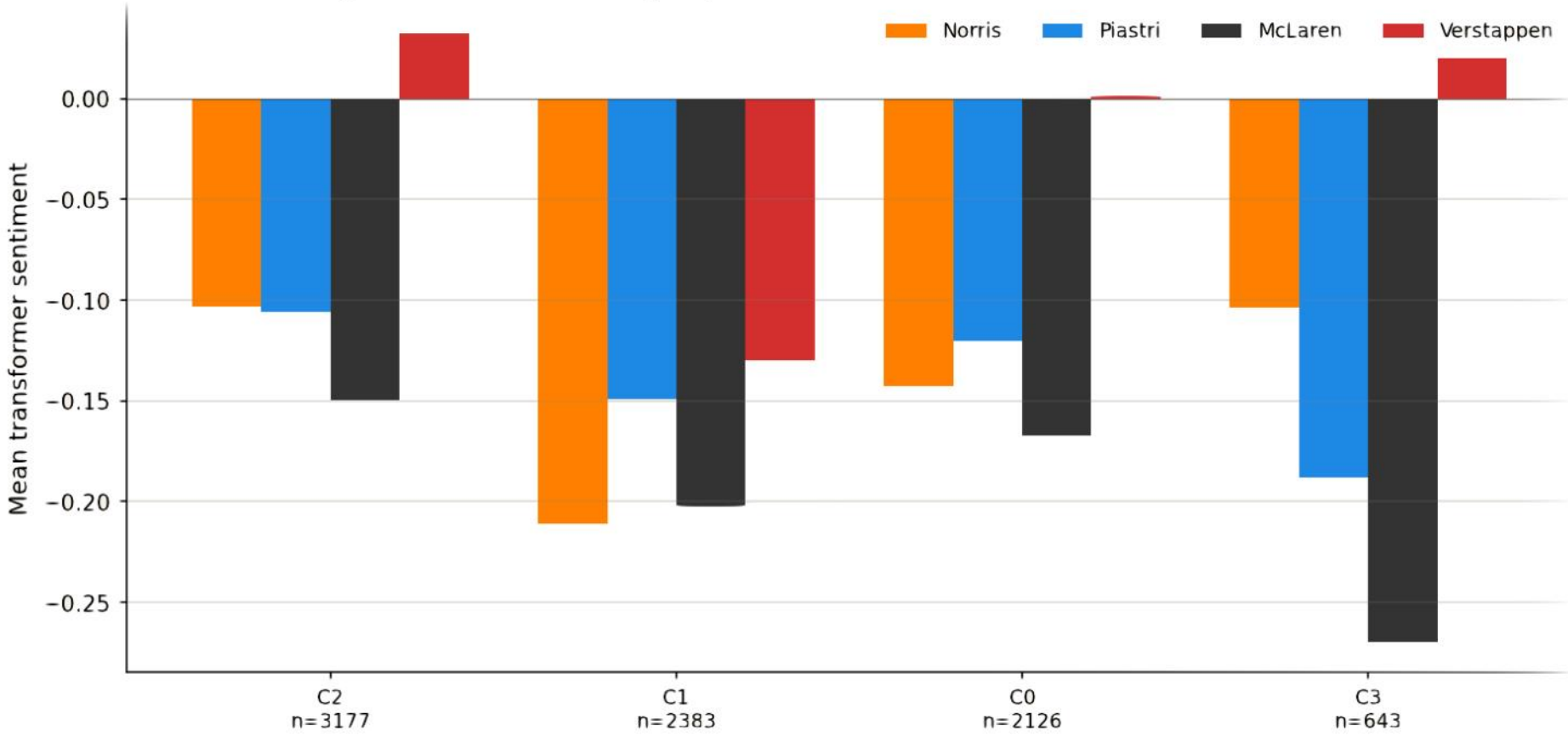
DID THE FANBASE SPLIT INTO TRIBES?

ANALYSIS 3 • LOUVAIN ON CO-ENGAGEMENT GRAPH • 8,329 USERS • 937,864 EDGES • MODULARITY 0.267

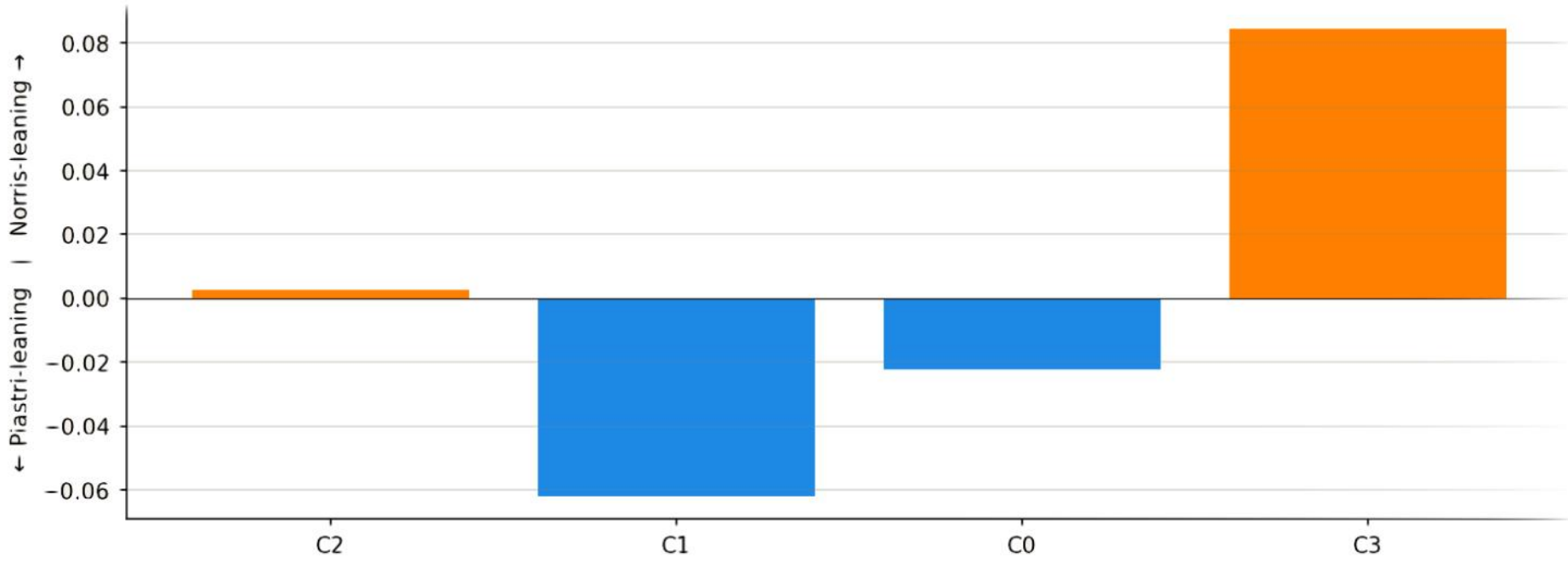
Commenter co-engagement network
Top 1,500 of 8,329 users by degree, colored by Louvain community



Per-community mean sentiment by aspect



Community polarization
Norris sentiment - Piastri sentiment



C3

Smallest community (n=643) is mildly Norris-leaning and holds the most negative McLaren-team sentiment (-0.27). The polarization is primarily emotional, not structural.

HOW WE MEASURED IT.

METHODS & STRATEGIES · OPEN-SOURCE PYTHON STACK

01

COLLECTION

Reddit's public JSON API. Polite rate-limiting, full comment-tree extraction, checkpoint resumability.

requests · python-stdlib

02

SENTIMENT

Transformer model (cardiffnlp/twitter-roberta), validated against VADER baseline at Pearson $r=0.44$.

transformers · torch

03

TOPICS

Latent Dirichlet Allocation, $K=5$ per event. TF-IDF features, n-gram (1,2), domain stop-word filtering.

scikit-learn · pandas

04

NETWORK

Bipartite users $\times$ posts $\rightarrow$ user-user co-engagement. Louvain community detection plus centrality.

networkx · python-louvain

05

ASPECTS

Regex-based entity-mention detection. Five aspects: Norris, Piastri, McLaren, Verstappen, Team mgmt.

re · pandas

06

VALIDATION

Two-method triangulation. VADER's documented positive bias justified the transformer as primary.

scipy · stratified sampling

07

EVENTS

Ten flashpoint races as analytical anchors. ± 7 -day windows for event-conditioned analyses.

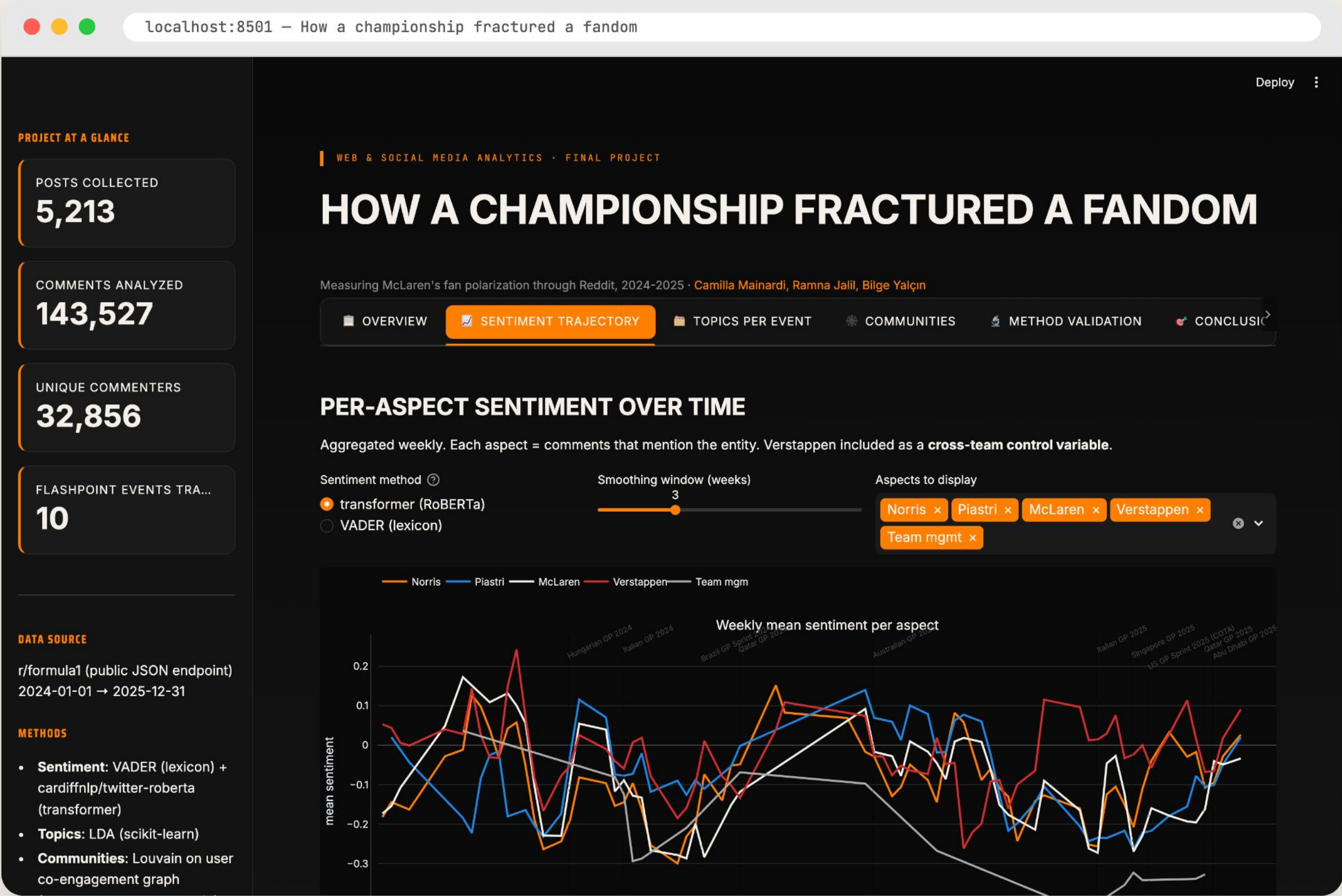
F1-expert sourced timeline

08

CONTROL

Verstappen sentiment as cross-team control variable to isolate McLaren-specific effects.

cross-aspect comparison



INTERACTIVE TRAJECTORY

Toggle aspects, switch VADER ↔ transformer, choose smoothing window.

EVENT DRILL-DOWN

Pick any of 10 flashpoints → top LDA topics with representative comments.

COMMUNITY EXPLORER

Per-community sentiment table + network image + bridge users ranked.

METHOD VALIDATION

VADER vs Transformer scatter, with disagreement examples surfaced.





A MULTI-LAYERED FRACTURE.

LAYER 01

TEMPORAL

McLaren-aspect sentiment crashes at every flashpoint while Verstappen baseline holds. Patient zero: Hungary 2024.

LAYER 02

TOPICAL

Distinct grievance topics emerge per event. 'Slow pit stops', 'Zak Brown', 'papaya rules' become discrete LDA clusters.

LAYER 03

STRUCTURAL

Mild Norris-leaning sub-tribe (n=643) holds the worst team sentiment. Polarization is detectable in the network.

"Winning didn't heal the fracture — and championship victories do not erase the brand debt."

